Supplementary Information: Programmable Quantum Resource Manifolds via Entanglement-Dynamic Topology

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Overview

This combined Supplementary Information accompanies the main-v4 manuscript, *Programmable Quantum Resource Manifolds via Entanglement-Dynamic Topology*. It merges the concise Overleaf supplement with the extended local SI material atlas. The central microscopic quantity is the curvature of electronic energies with respect to a phonon normal-mode coordinate,

$$\kappa_{n\alpha}(k) = \frac{\partial^2 E_n(k, Q_\alpha)}{\partial Q_\alpha^2}, \tag{1}$$

which captures the momentum-resolved second-order response of electronic structure to lattice motion. This curvature encodes non-adiabatic electron–phonon coupling and becomes strongly structured near degeneracies, where electronic states form coupled resource manifolds [1, 2].

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S1. Curvature Derivation and Calculation Protocol

Starting from the electron–phonon Hamiltonian,

$$H = H_e(\mathbf{r}; \{Q_\alpha\}) + \sum_\alpha \left(\frac{P_\alpha^2}{2M_\alpha} + \frac{1}{2} M_\alpha \omega_\alpha^2 Q_\alpha^2 \right), \tag{2}$$

expanding the electronic Hamiltonian to second order in the normal-mode coordinates gives

$$H_e(\{Q_\alpha\}) \approx H_e^{(0)} + \sum_\alpha Q_\alpha \partial_\alpha H_e + \frac{1}{2} \sum_{\alpha\beta} Q_\alpha Q_\beta \partial_\alpha \partial_\beta H_e. \tag{3}$$

Projection onto an adiabatic electronic basis introduces non-adiabatic coupling terms (NACTs),

$$A_{mn}^{(\alpha)} = \langle m | \partial_{Q_\alpha} | n \rangle, \tag{4}$$

which generate off-diagonal mixing and geometric phases [3, 4].

Electronic structure and electron–vibration interactions were computed using all-electron quantum solid-state methods with numerical atomic orbitals, with additional cross-checks against plane-wave density-functional calculations where available [**?**, **?**, **?**, **?**]. For each selected phonon normal mode Q_α, band energies were recalculated for symmetric positive and negative static displacements around the equilibrium geometry. Second-order derivatives were evaluated using the five-point finite difference stencil

$$\frac{\partial^2 A}{\partial u_\nu^2} \approx \frac{-A(-2\Delta u) + 16A(-\Delta u) - 30A(0) + 16A(\Delta u) - A(2\Delta u)}{12(\Delta u)^2}. \tag{5}$$

The plotted traces are therefore momentum resolved, but the derivative is taken with respect to the lattice normal-mode coordinate.

Near a quasi-degenerate two-band subspace $\{|i\rangle, |j\rangle\}$, the leading interband-mixing contribution has the perturbative form

$$\left. \frac{\partial^2 E_i}{\partial Q_\alpha^2} \right|_{\text{mix}} \approx + \frac{2|g_{ij}^{(\alpha)}(k)|^2}{\Delta_{ij}(k)}, \tag{6}$$

$$\left. \frac{\partial^2 E_j}{\partial Q_\alpha^2} \right|_{\text{mix}} \approx - \frac{2|g_{ij}^{(\alpha)}(k)|^2}{\Delta_{ij}(k)}, \tag{7}$$

where $\Delta_{ij}(k) = E_i(k) - E_j(k)$ and $g_{ij}^{(\alpha)}(k) = \langle i | \partial H / \partial Q_\alpha | j \rangle$. The equal-and-opposite signs express trace conservation within the local two-level subspace. The associated NACT is

$$A_{ij}^{(\alpha)}(k) = \langle i | \partial_{Q_\alpha} | j \rangle = \frac{g_{ij}^{(\alpha)}(k)}{E_j(k) - E_i(k)}, \qquad i \neq j, \tag{8}$$

so curvature peaks are expected where the local gap is small and symmetry allows the selected mode to couple the partner branches.

For the alkali-metal reference workflow [5], longitudinal modes along symmetry-selected directions produced the strongest antisymmetric curvature response, while transverse or mixed-polarized modes generally produced weaker and more symmetric modulation. In the present EDT analysis, paired sign-opposite curvature branches are treated as evidence of local non-adiabatic mixing, and the subsequent window extraction tests whether that mixing remains smooth enough to support a finite quadratic manifold.

When a displacement amplitude is needed to convert curvature into an energy modulation, the second-order correction is

$$\Delta E_n(k) \approx \frac{1}{2}\kappa_{n\alpha}(k)(\Delta Q_\alpha)^2. \tag{9}$$

For zero-point or low-temperature estimates, ΔQ_α may be obtained from the Debye–Waller amplitude or the corresponding harmonic-oscillator normal-mode amplitude [**?**, **?**]. The archived K_3C_{60} workflow used here starts from already-computed curvature traces, so the finite-window analysis does not reconstruct the displaced geometries.

S2. EDT Window Algorithm and K_3C_{60} Details

To identify finite EDT windows from the electronic-structure data, we analyze paired curvature branches near a local band-crossing region. For each selected branch pair, the curvature response is approximated locally by

$$\kappa_\pm(k) \approx a_\pm(k - k_0)^2 + b_\pm, \tag{10}$$

where k_0 denotes the local center of the paired response and $\kappa_\pm(k)$ are the two curvature branches. The EDT window Δk_{EDT} is then defined as the maximal contiguous momentum interval over which the local quadratic description remains valid within a prescribed normalized residual threshold. For the K_3C_{60} case study, the analysis uses an archived near-Fermi band-context figure together with a mode-resolved second-order band-curvature dataset. The representative phonon mode is $m120$, with frequency 1090.35 cm^{-1}. The curvature dataset contains precomputed traces of

$$\frac{\partial^2 E_n(k, Q_\alpha)}{\partial Q_\alpha^2}$$

for multiple band indices along the Γ–X–L–W–Γ path. Thus, the EDT extraction is performed directly on archived second-order curvature data, rather than by reconstructing displaced structures or recomputing finite-difference derivatives of the band energies.

The main analysis focuses on the paired near-Fermi branches labeled Bands 208 and 209.

A local momentum interval surrounding the crossing region is selected, and each branch is centered at its strongest local curvature extremum: the negative branch at its local minimum and the positive branch at its local maximum. Around each extremum, a symmetric neighborhood is used to obtain an independent quadratic fit. The resulting parabolic approximations define the local curvature model for the two branches.

To quantify the validity of this local model, we compute a branch-resolved normalized residual,

$$R_{\pm}(k) = \frac{\left|\kappa_{\pm}(k) - \kappa_{\pm}^{\mathrm{fit}}(k)\right|}{\max_{k\in\mathcal{I}} |\kappa_{\pm}(k)|}, \tag{11}$$

where $\mathcal{I}$ is the selected momentum interval and $\kappa_{\pm}^{\mathrm{fit}}(k)$ denotes the fitted quadratic form. For a nominal threshold $R_{\mathrm{th}} = 0.08$, the valid region for each branch is defined as the contiguous interval connected to its curvature extremum over which $R_{\pm}(k) < R_{\mathrm{th}}$. The EDT window Δk_{EDT} is defined as the union of the two branch-valid intervals, while the core window Δk_{core} is defined as their overlap when a nonempty overlap exists. In this way, Δk_{EDT} measures the full finite region over which the paired curvature structure remains locally stable, whereas Δk_{core} identifies the stricter interval over which both branches simultaneously satisfy the EDT criterion.

Robustness of the extracted window is assessed by varying the residual threshold, the size of the local fitting neighborhood, the selected neighboring band pair, and the effective k-grid resolution. In addition, bootstrap resampling of the local fit points is used to estimate the uncertainty in Δk_{EDT}. These tests verify that the finite curvature window is not an artifact of a single fitting choice or threshold value, but persists under reasonable variations of the extraction procedure.

Paired curvature branches near crossings are fit locally to

$$\kappa_{\pm}(k) \approx a_{\pm}(k - k_0)^2 + b_{\pm}. \tag{12}$$

The EDT window Δk_{EDT} is defined as the interval over which the local quadratic residual remains below a prescribed threshold. This gives a reproducible, data-driven criterion for identifying finite structured manifolds.

S3. Expanded Curvature-Manifold Materials Study

To test whether the curvature-window diagnostic used for K_3C_{60} is specific to a single material example, we apply the same workflow to representative branch pairs from elemental V, elemental Na, LaH_{10}, and $YBa_2Cu_3O_7$. These systems provide a small but diverse control set, including a vairous superconducting and elemental materials. For each material, we use the same paired-curvature analysis, local quadratic fitting, residual-threshold criterion, and bootstrap uncertainty estimate as in the main-text K_3C_{60} analysis. The cross-material

results show that EDT-like curvature windows can be extracted for all representative materials considered here, but the size of the window depends strongly on material complexity, symmetry, and branch character. In the simpler metallic systems, such as V and Na, the selected curvature branches remain compatible with a local quadratic description over a broader momentum range. This produces wider EDT-like windows and suggests that the relevant curvature structure is more extended in momentum space. By contrast, in the more complex materials, such as $YBa_2Cu_3O_7$ and LaH_{10}, the curvature response is more localized, so the residual criterion is satisfied only over a narrower interval. In these cases, EDT-like organization is concentrated into a more focused momentum-space region. Figure S1 summarizes this trend. The bar heights report the nominal extracted window widths, while the error bars show the sensitivity of those widths to bootstrap refitting of the local quadratic branch structure. IThis comparison shows that the curvature-window diagnostic is transferable across materials, while also revealing that the extent of the EDT-like regime reflects the underlying material class and curvature organization.

S4. Threshold, Bootstrap, and Temperature Robustness

Figure S11 collects the analysis variations used to test the stability of the K_3C_{60} curvature-defined EDT window. The nominal configuration uses Bands 208 and 209 of mode `m120`, analysis indices 150–281, local quadratic fit half-window of 18 points, and residual threshold $R_{\mathrm{th}} = 0.08$. The resulting nominal width is $\Delta k_{\mathrm{EDT}} = 0.154$.

A residual-threshold sweep over $R_{\mathrm{th}} \in [0.04, 0.18]$ yields widths between 0.137 and 0.178. A fit-window sweep over half-windows from 10 to 28 points yields widths between 0.127 and 0.199. A local-fit bootstrap around the nominal paired branches gives a 95% confidence interval of $[0.147, 0.158]$ for the extracted width. Subsampling the k grid changes the width only modestly.

Temperature-dependent band structures are treated as a dynamical probe of NACT-induced geometry. In EDT-compatible systems such as Nb, temperature-induced renormalization preserves band connectivity and near-degenerate structure, maintaining paired smooth curvature profiles. In sharper systems, temperature can enhance nonuniform degeneracy lifting and reduce the stability of a local quadratic description.

S5. Curvature Topology Classification

Beyond magnitude, the topology of curvature near degeneracies provides an additional diagnostic for classifying electron–vibration coupling regimes. Here curvature is defined with respect to a normal-mode coordinate and evaluated along a momentum path [1, 2]. The purpose of this classification is not to replace the branch-resolved EDT-window extraction, but

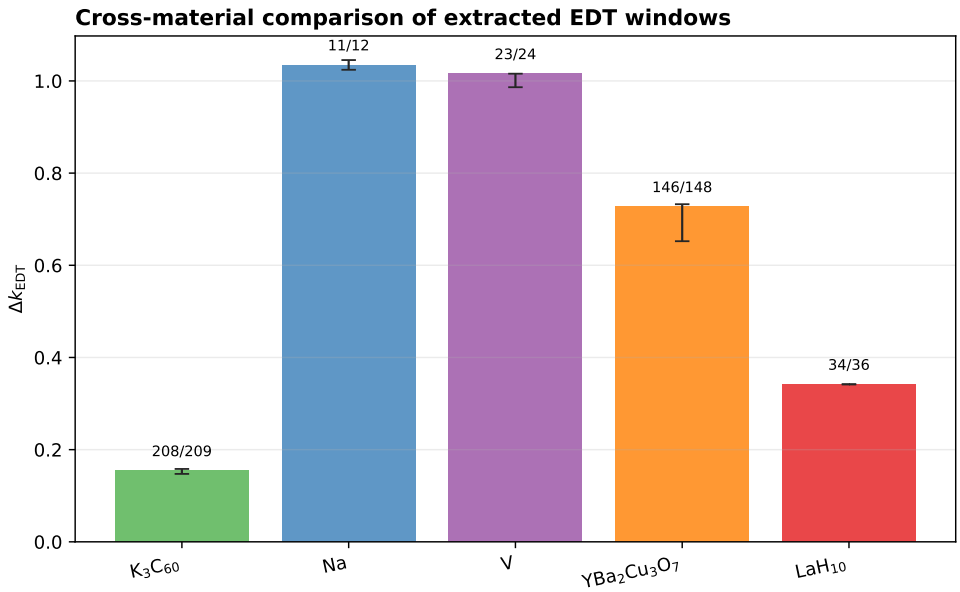


Figure S1: **Cross-material summary of extracted EDT-window widths.** The bar heights show the residual-defined Δk_{EDT} values extracted from the selected representative branch pairs in K_3C_{60}, Na, V, $YBa_2Cu_3O_7$, and LaH_{10}, while the error bars show bootstrap confidence intervals from the local quadratic-fit procedure. This compact summary is the supplementary overview of the branch-resolved analyses shown in the figures below and highlights that the same curvature-window diagnostic does not produce one universal window scale across materials.

Table S1: Representative branch-pair results extracted from the expanded curvature-manifold workflow at the common residual threshold $R_{\mathrm{th}} = 0.08$. Widths are reported in the native k coordinate of each archived export, and the bootstrap interval corresponds to the 95% width confidence range from the replicated local fit.

Material	Pair	k_0	Δk_{EDT}	Bootstrap 95% CI
K_3C_{60}	208/209	0.823	0.154	[0.147, 0.158]
V	22/23	2.718	1.016	[0.986, 1.016]
Na	11/12	1.837	1.035	[1.024, 1.045]
$YBa_2Cu_3O_7$	145/147 subset	1.599	0.727	[0.652, 0.733]
LaH_{10}	33/35 subset	1.300	0.342	[0.342, 0.342]

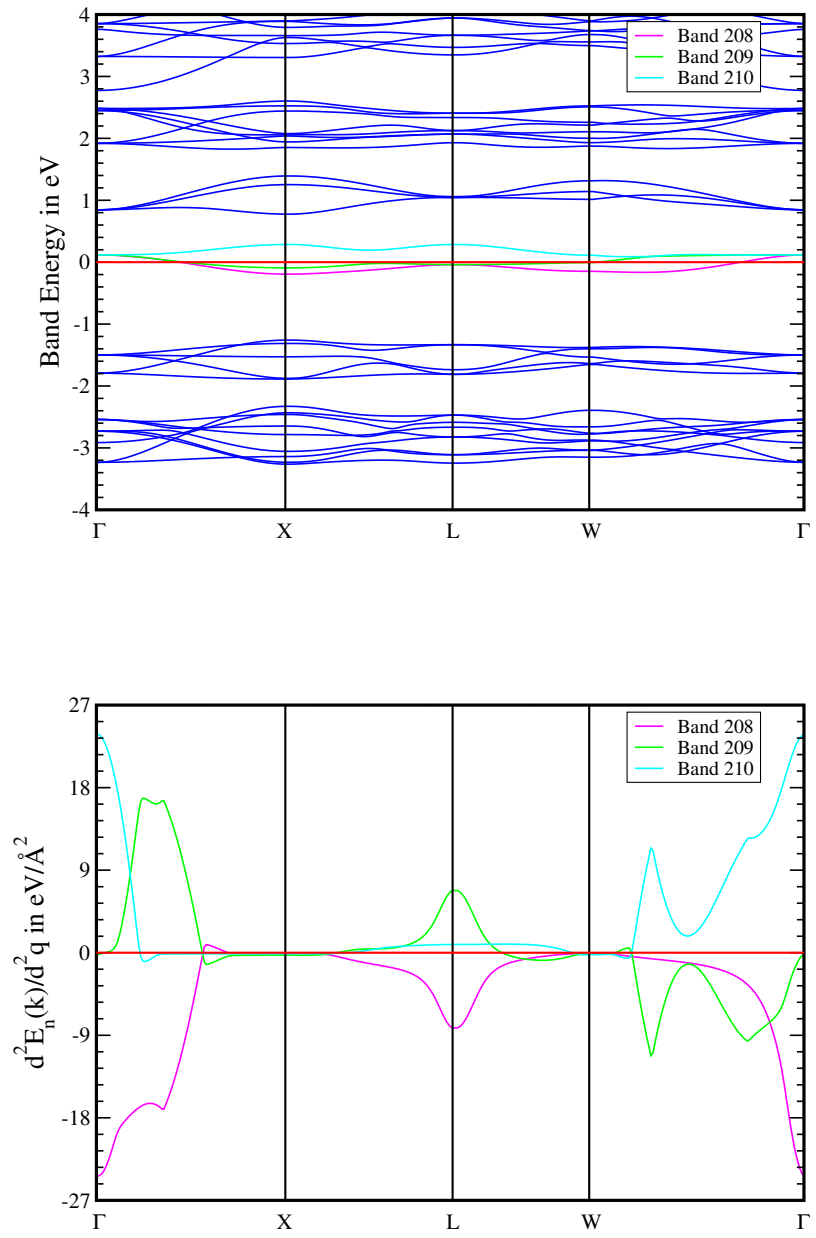


Figure S2: **Curvature-resolved electronic structure of K_3C_{60}. Top,** Electronic band structure of cubic K_3C_{60}, showing multiple flat, near-degenerate bands in the vicinity of the Fermi level. **Bottom,** Second-order derivatives of band energies with respect to a normal-mode coordinate ($\partial^2 E_n/\partial Q_\alpha^2$) for a representative phonon mode at frequency 1090.35 cm^{-1}. Pronounced curvature amplitudes are observed near band crossings and degeneracies, indicating strong non-adiabatic electron–phonon coupling. These structured curvature features define a broad EDT window, in which lattice-induced modulation organizes electronic states into coupled manifolds.

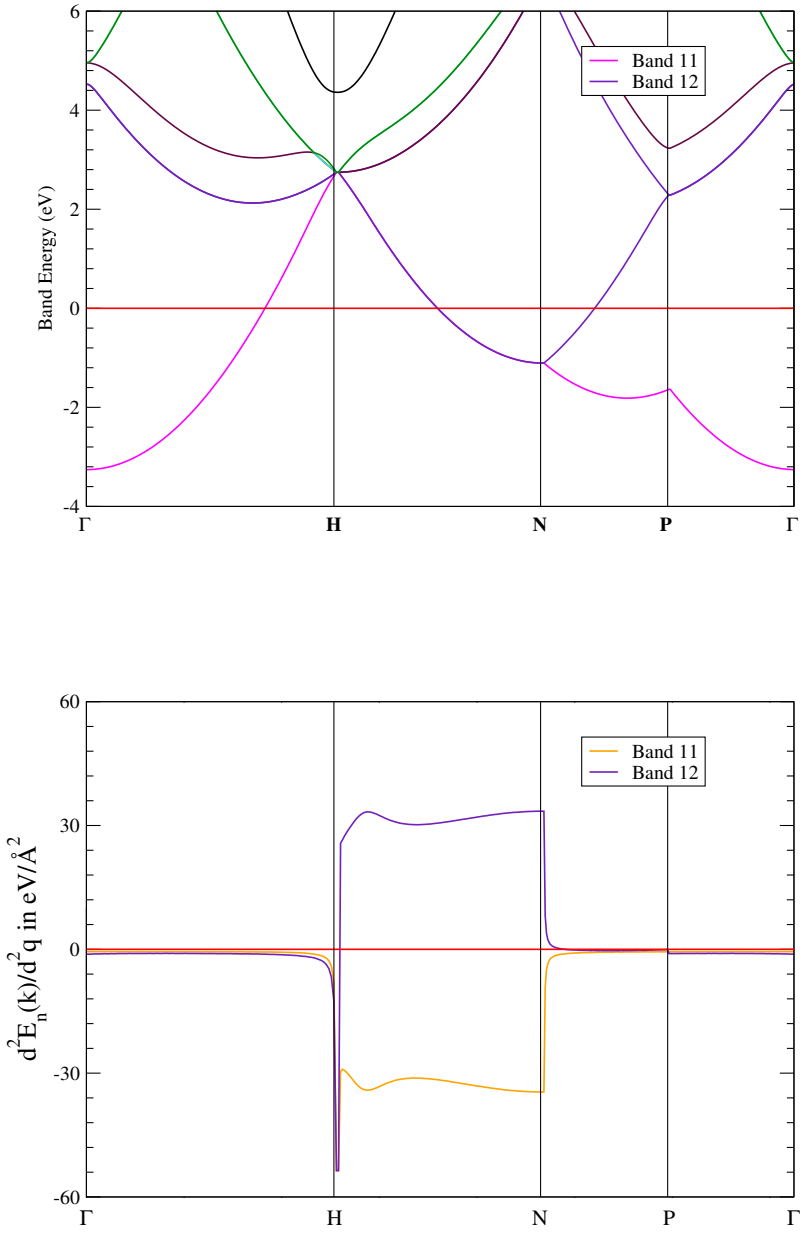


Figure S3: **Curvature-resolved electronic structure of Na. Top,** Electronic band structure of body-centered cubic (bcc) Na along high-symmetry directions, showing simple, nearly free-electron-like dispersion with weak band interactions near the Fermi level. **Bottom,** Second-order derivatives of band energies with respect to a normal-mode coordinate ($\partial^2 E_n/\partial Q_\alpha^2$) of bands 11 and 12 for a representative phonon mode at frequency 117.39 cm^{-1}. The curvature exhibits a monotonic profile without nodes, indicating a single-sign second-order response across the momentum range. This reflects minimal electronic mixing under lattice motion and the absence of paired curvature structure. As a result, no local quadratic manifold can be defined, and no EDT window is observed.

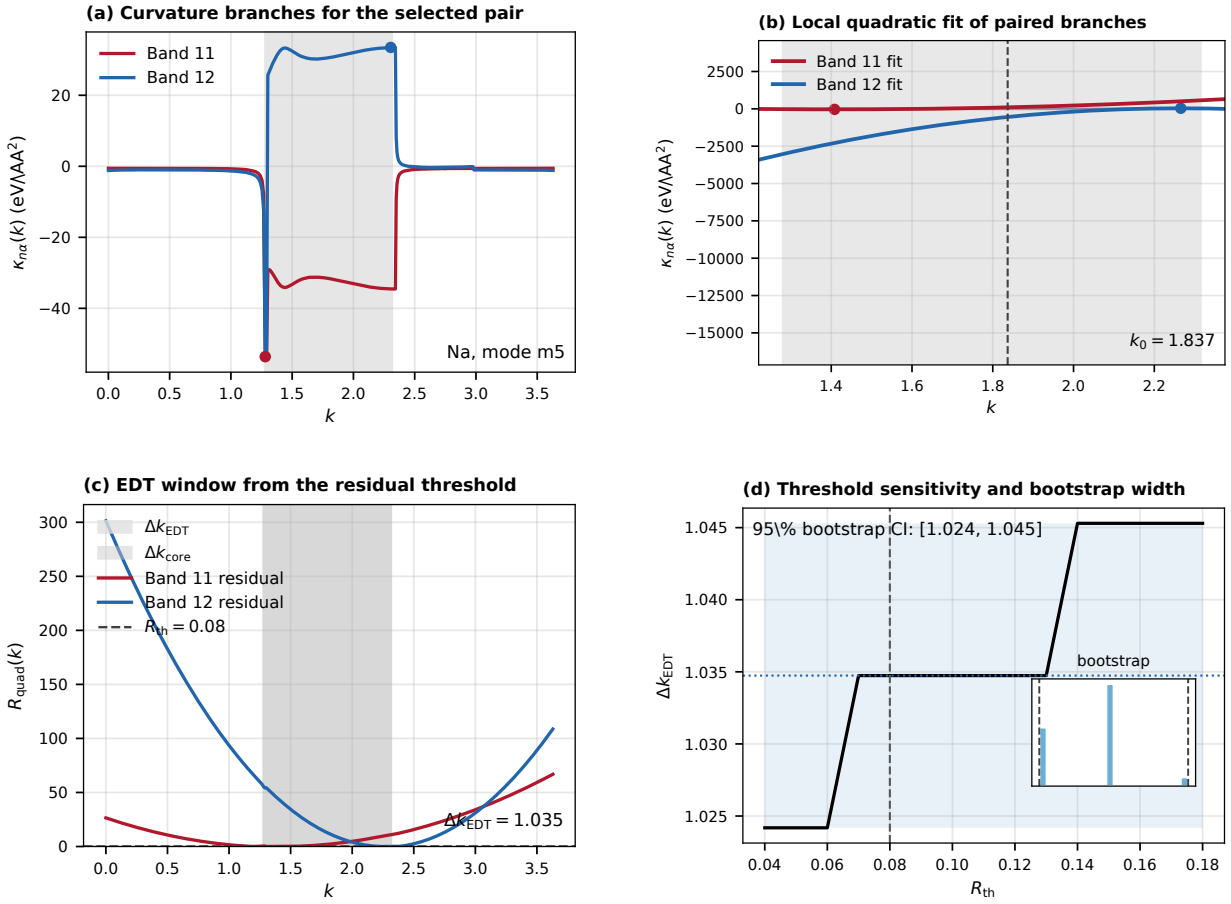


Figure S4: **Extended curvature-manifold analysis for Na.** (b) Selected paired curvature branches for the archived mode-resolved export. (c) Local quadratic fit of the paired branches. (d) Residual-defined candidate EDT window at the common threshold $R_{\mathrm{th}} = 0.08$. (e) Threshold sensitivity and bootstrap width summary. Na provides the simple-metal baseline for the expanded materials workflow using Bands 11 and 12, with a representative window centered at $k_0 \approx 1.84$ and $\Delta k_{\mathrm{EDT}} \approx 1.035$.

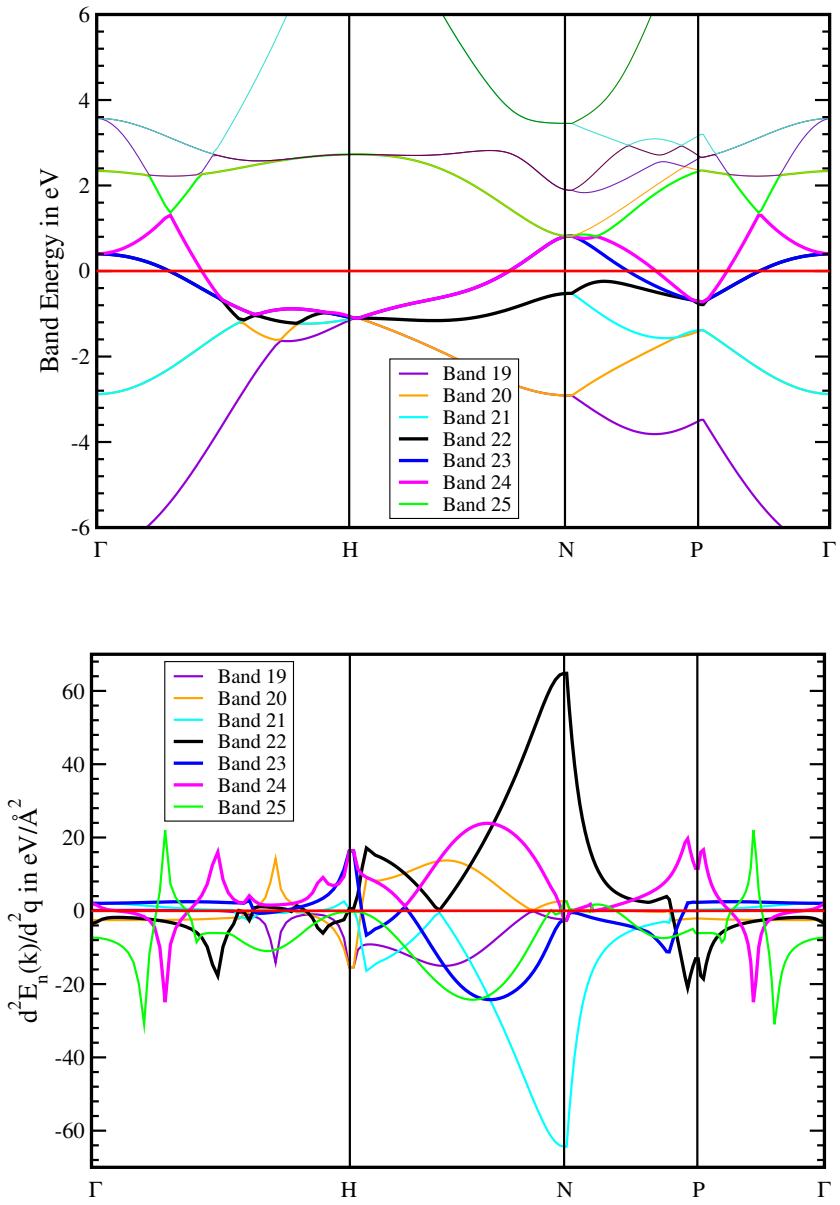


Figure S5: **Curvature-resolved electronic structure of V. Top,** Electronic band structure of bcc V along high-symmetry points, showing multiple near-Fermi band crossings. **Bottom,** Second-order derivatives of band energies with respect to a normal-mode coordinate ($\partial^2 E_n/\partial Q_\alpha^2$) of bands 19–25 for a representative phonon mode at frequency 260.98 cm^{-1}. The curvature exhibits paired two-node structures connected by smooth, slowly varying branches, characteristic of coherent multi-band coupling. In contrast to simple metals, these curvature profiles admit a consistent local quadratic description over a finite momentum interval, defining a well-resolved EDT window.

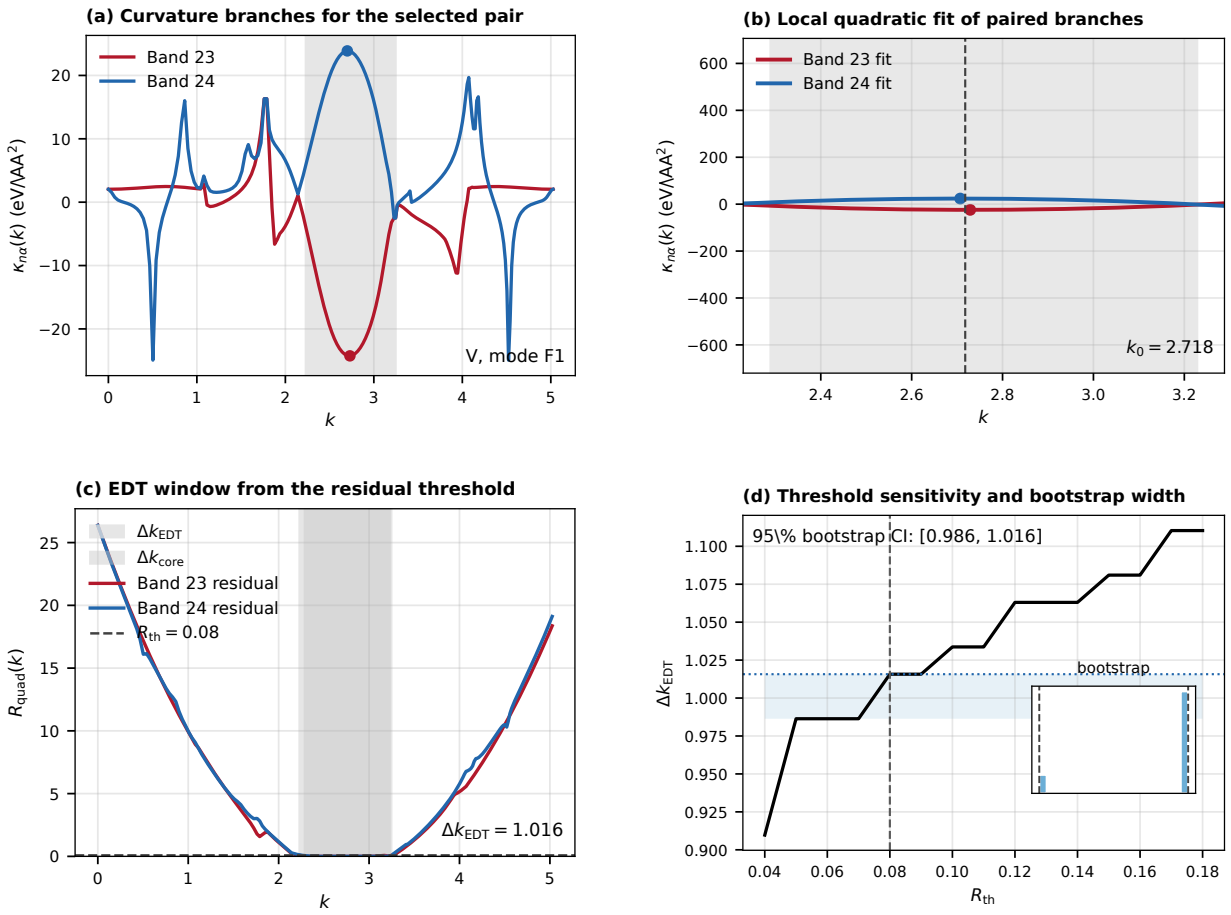


Figure S6: **Extended curvature-manifold analysis for V.** (b) Selected paired curvature branches for the archived mode-resolved export. (c) Local quadratic fit of the paired branches. (d) Residual-defined candidate EDT window at the common threshold $R_{\mathrm{th}} = 0.08$. (e) Threshold sensitivity and bootstrap width summary. V provides the broadest representative window in the expanded study, with $k_0 \approx 2.72$ and $\Delta k_{\mathrm{EDT}} \approx 1.02$.

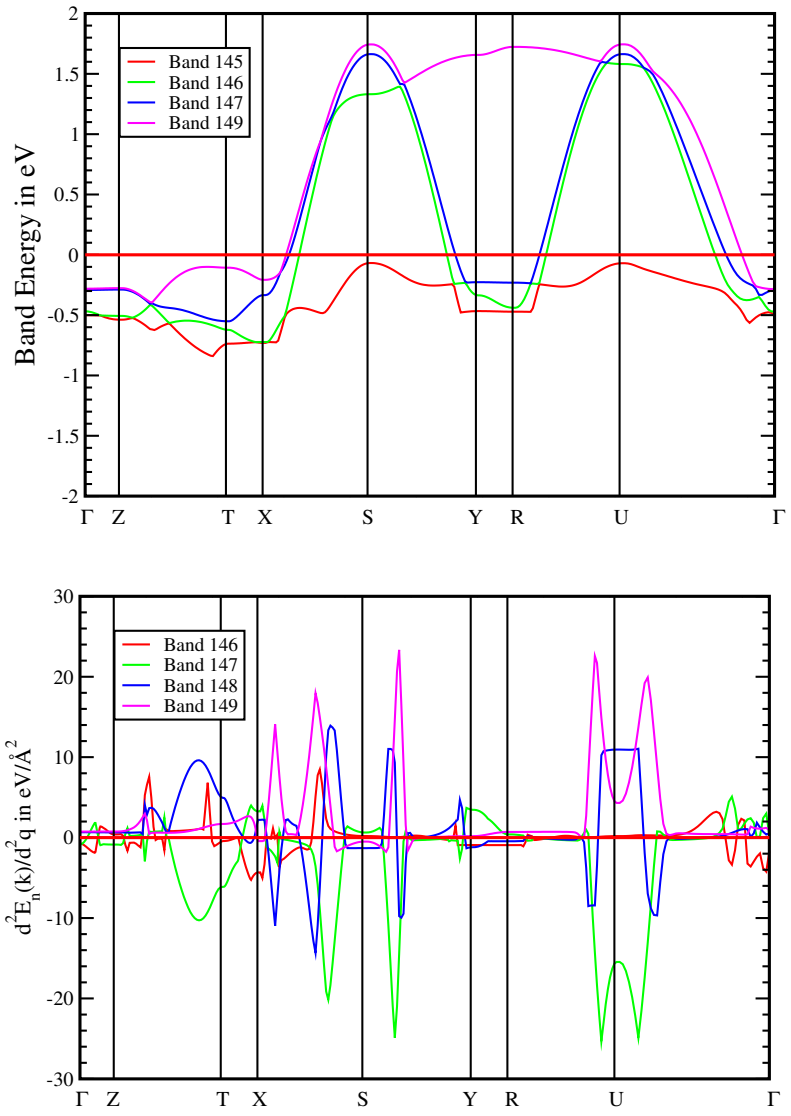


Figure S7: **Curvature-resolved electronic structure of $YBa_2Cu_3O_7$. Top,** Electronic band structure of $YBa_2Cu_3O_7$ along high-symmetry directions, showing multiple near-Fermi band crossings. **Bottom,** Second-order derivatives of band energies with respect to a normal-mode coordinate ($\partial^2 E_n/\partial Q_\alpha^2$) of bands 146–149 for a representative phonon mode at frequency 439.72 cm^{-1}. The curvature exhibits two-node structures associated with band degeneracies, but is characterized by sharp, localized peaks and rapidly varying behavior.

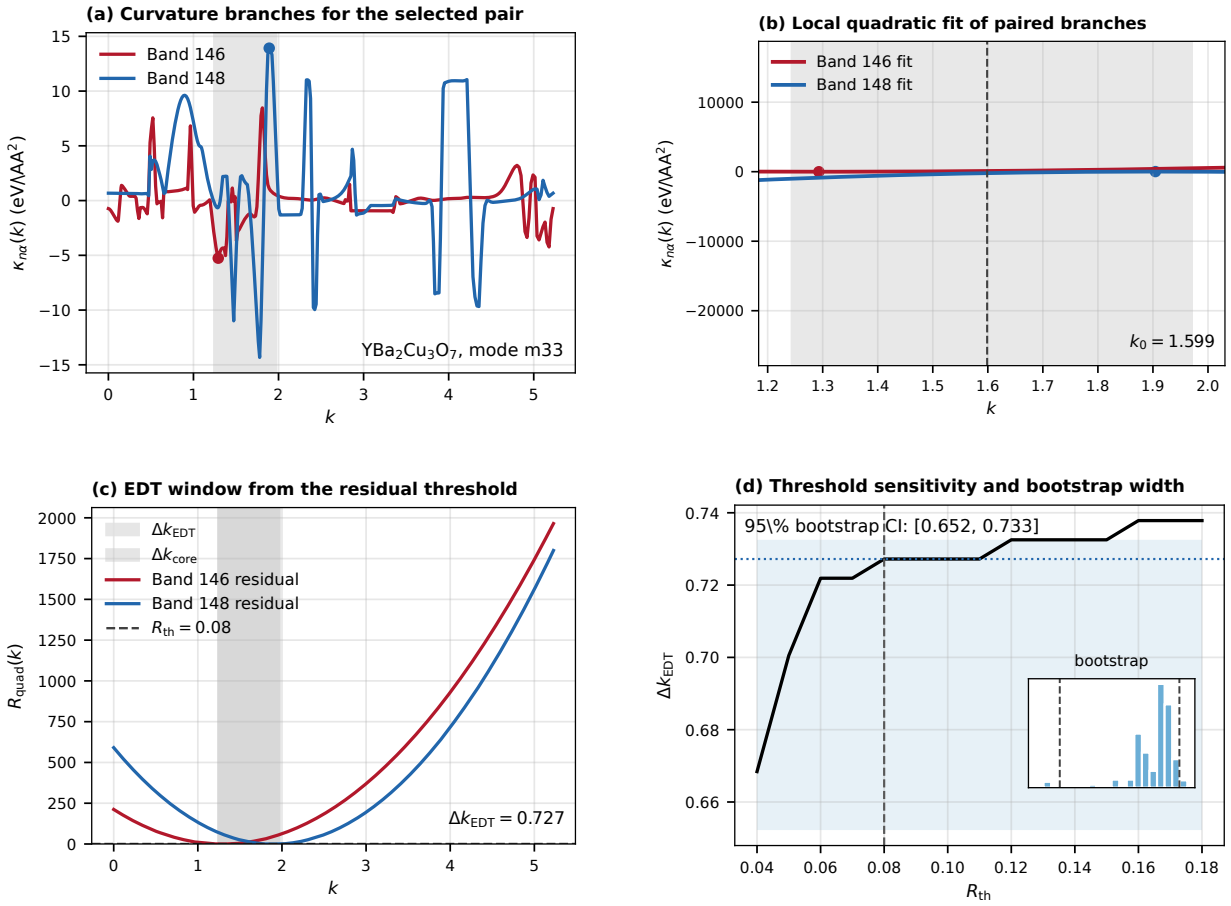


Figure S8: **Extended curvature-manifold analysis for $YBa_2Cu_3O_7$.** (b) Selected paired curvature branches for the archived mode-resolved export. (c) Local quadratic fit of the paired branches. (d) Residual-defined candidate EDT window at the common threshold $R_{\mathrm{th}} = 0.08$. (e) Threshold sensitivity and bootstrap width summary. The representative pair shown here is drawn from the requested 145/146/147/149 band subset and gives a localized window around $k_0 \approx 1.60$ with $\Delta k_{\mathrm{EDT}} \approx 0.727$.

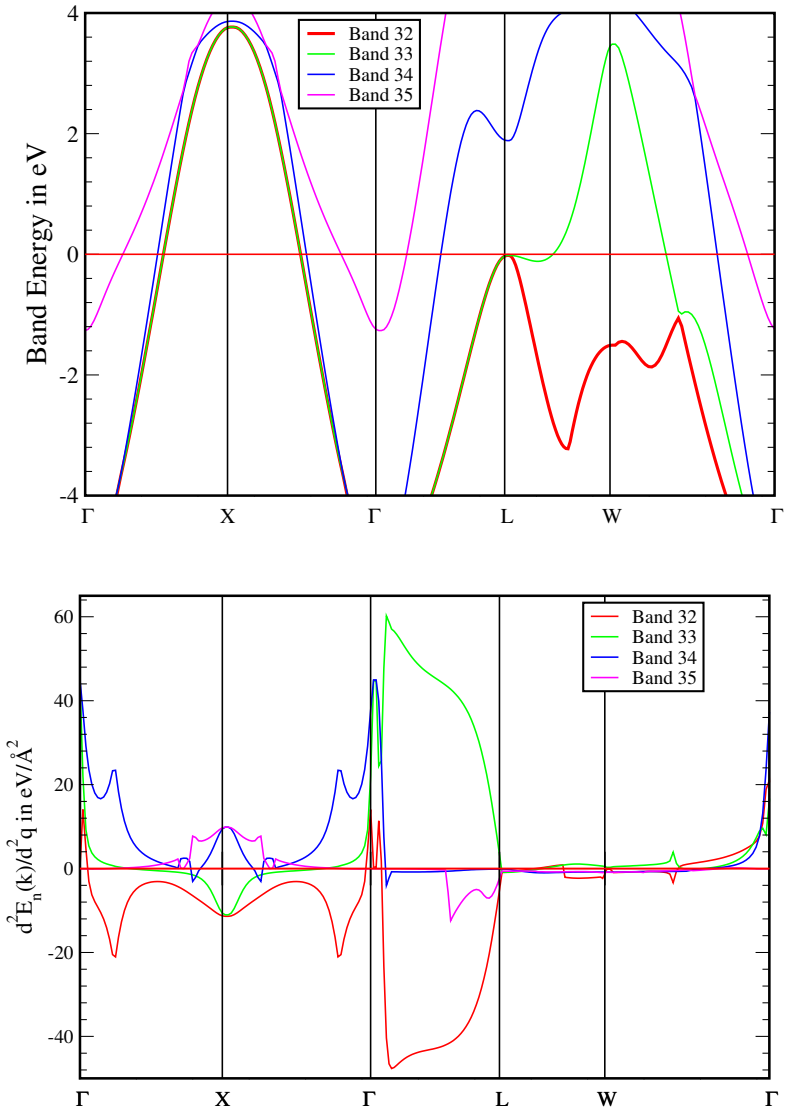


Figure S9: **Curvature-resolved electronic structure of LaH_{10}. Top,** Electronic band structure of cubic LaH_{10} at high pressure (300 GPa), showing multiple near-degenerate bands in the vicinity of the Fermi level. **Bottom,** Second-order derivatives of band energies with respect to a normal-mode coordinate ($\partial^2 E_n/\partial Q_\alpha^2$) for a representative phonon mode at frequency 1137.54 cm^{-1}. Large curvature amplitudes are observed near band crossings and degeneracies, indicating strong non-adiabatic electron–phonon coupling. These structured curvature features define an extended EDT regime, in which lattice-induced modulation organizes electronic states into coupled manifolds.

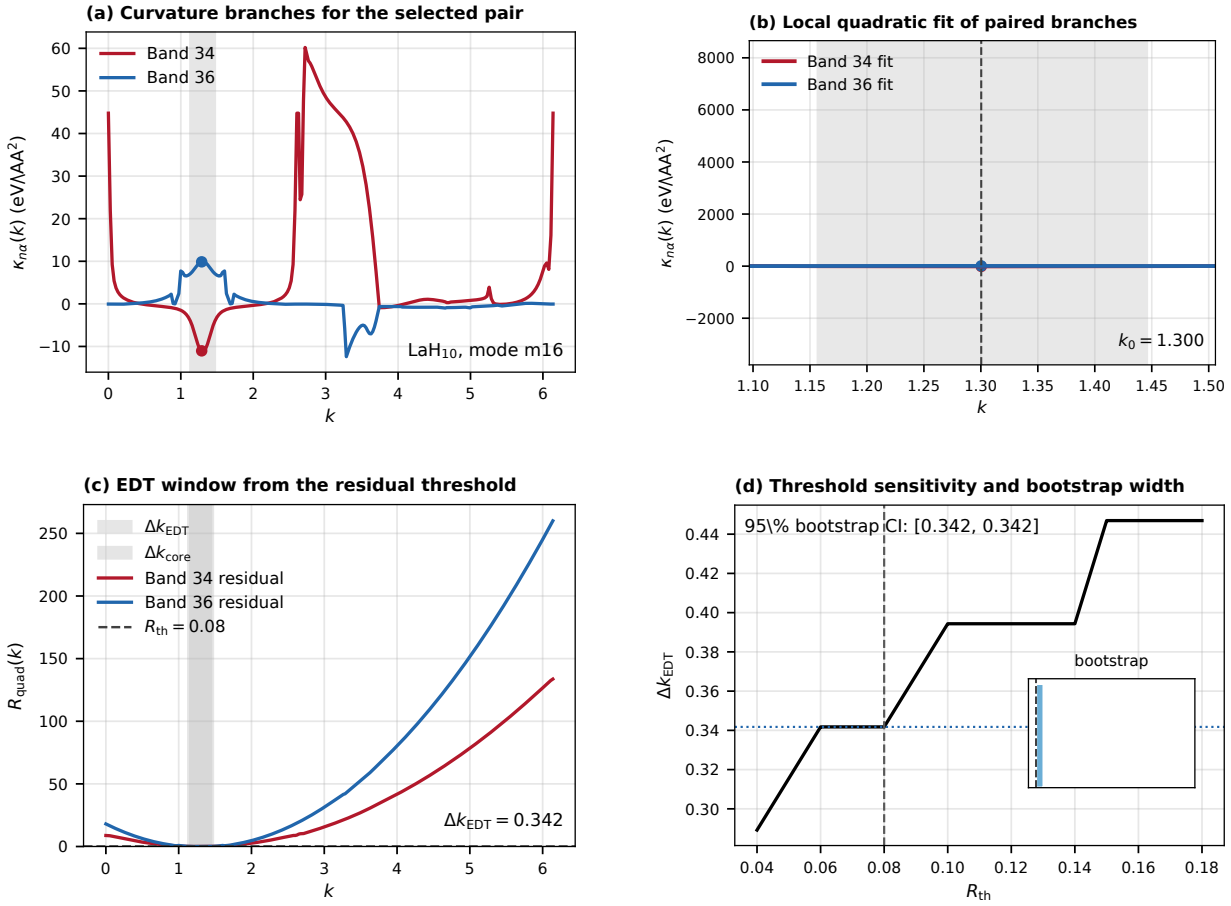


Figure S10: **Extended curvature-manifold analysis for LaH_{10}.** (b) Selected paired curvature branches for the archived mode-resolved export. (c) Local quadratic fit of the paired branches. (d) Residual-defined candidate EDT window at the common threshold $R_{\mathrm{th}} = 0.08$. (e) Threshold sensitivity and bootstrap width summary. The representative pair shown here is drawn from the 32–35 band subset and yields a localized window near $k_0 \approx 1.30$ with $\Delta k_{\mathrm{EDT}} \approx 0.342$.

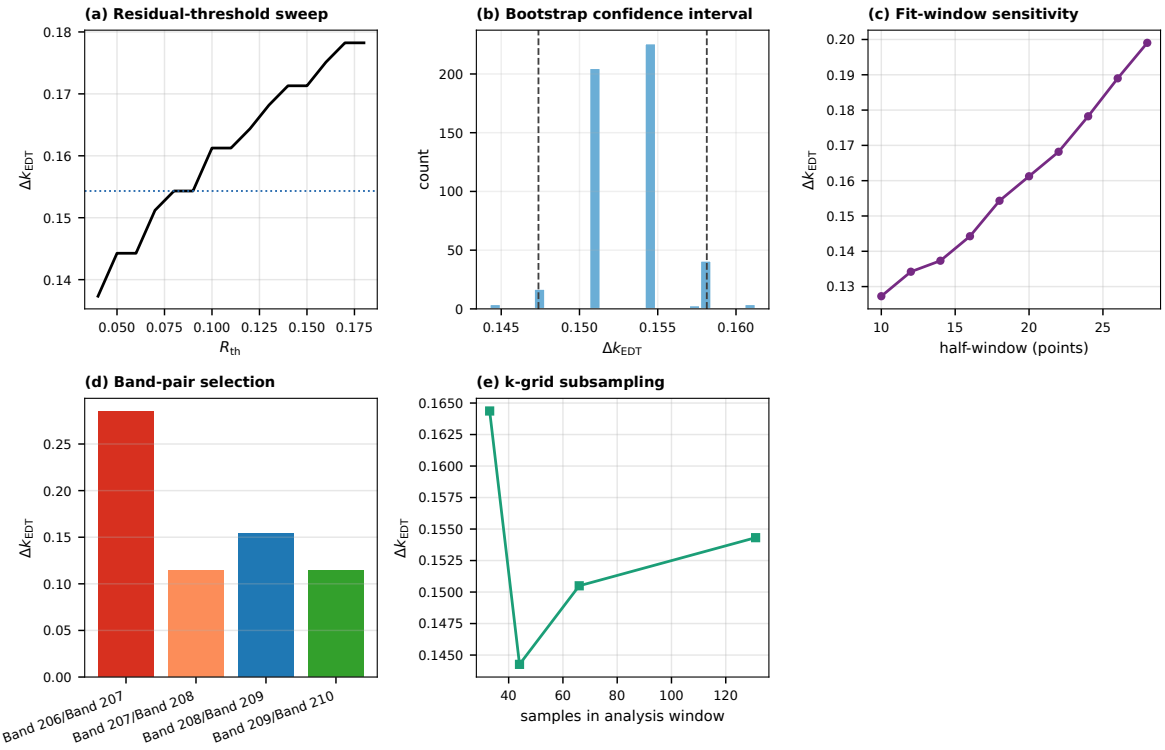


Figure S11: **Robustness checks for the K_3C_{60} curvature-defined EDT window.** The panels summarize residual-threshold sensitivity, bootstrap uncertainty, local-fit-window sensitivity, neighboring band-pair choices, k-grid decimation, and the unavailable displacement-amplitude sweep in the bundled archive.

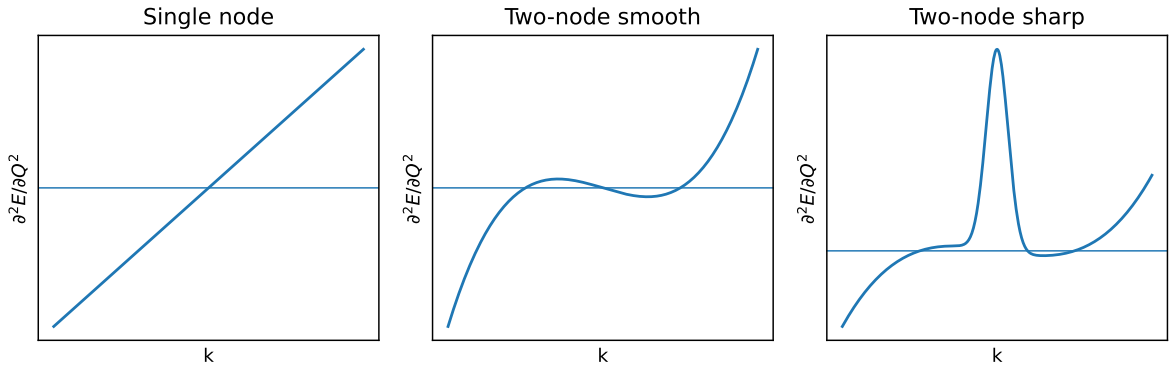


Figure S12: **Schematic classification of curvature topology.** Representative curvature profiles $\partial^2 E/\partial Q_\alpha^2$ illustrating three regimes: **left,** single-node curvature typical of simple metals; **middle,** two-node smooth curvature with slowly varying parabolic branches supporting a finite EDT window; and **right,** two-node sharp curvature with localized peaks that suppress the formation of a stable quadratic manifold. These classes are governed primarily by band dispersion and lattice-scale parameters rather than pairing symmetry alone.

to provide a higher-level geometric language for understanding why some materials support broad, smoothly varying finite windows while others remain sharper and more localized.

Single-node curvature. In alkali and simple metallic systems, the curvature often exhibits a single node or monotonic profile. This reflects relatively simple band dispersion and limited mixing in the relevant subspace. In that case, the second-order response changes sign at most once or remains one-sided, so there is little evidence for a paired branch structure capable of supporting a stable local quadratic manifold.

Two-node smooth curvature. In systems such as Nb and K_3C_{60}, paired bands near the Fermi level produce two nodes connected by smooth, approximately parabolic branches. The curvature varies slowly across a finite region, allowing a consistent local quadratic description and yielding a finite EDT window. This is the topology most naturally associated with EDT-compatible behavior, because the curvature is structured enough to indicate strong multi-band organization, yet smooth enough to remain fit-compatible over a nonzero momentum interval.

Two-node sharp curvature. In systems such as $YBa_2Cu_3O_7$ and LaH_{10}, two nodes can be present but connected by sharply varying curvature with localized peaks. This rapid variation prevents stable quadratic fitting despite strong coupling. The key point is that strong electron–vibration response alone is not sufficient: if the curvature changes too abruptly, the local manifold becomes too fragmented to support a stable quadratic window.

This topology view helps interpret the updated materials atlas in Sec. S3. The Na

and V selections occupy the single-node to smooth-two-node side of the classification and therefore support broader or more stable extracted windows. By contrast, the selected $YBa_2Cu_3O_7$ and LaH_{10} subsets sit closer to the sharp two-node regime, so their extracted windows remain more localized even when a finite branch-resolved fit is still possible. In that sense, Fig. S12 provides the qualitative geometric bridge between the raw curvature traces and the quantitative Δk_{EDT} comparison reported earlier in the supplement.

S6. Plaquette Model and Trotter Circuit

The minimal plaquette model used in the main text is

$$H = J \sum_{\langle i,j \rangle} Z_i Z_j + h \sum_i X_i + \lambda Z_0 Z_1 Z_2 Z_3. \tag{13}$$

This model isolates the ingredients needed for finite EDT dynamics: local couplings, transverse control that enables operator spreading, and a loop term that introduces nonlocal correlations. It is not presented as a quantitatively fitted Hamiltonian for K_3C_{60}.

The reproducible implementation is the package `src/plaquette_model`. The Hamiltonian constructor is `build_hamiltonian(J, h, lambda_loop, n_qubits=4)`. Exact state trajectories, OTOCs, two-point correlations, entanglement entropies, finite-regime scores, and Trotterized evolutions are exposed by the same package.

Time evolution under the minimal plaquette Hamiltonian is approximated using first-order Trotterization, with circuit depth and gate count scaling linearly with evolution time [6, 7, 8, 9]. Observable errors are quantified by comparing exact and Trotterized expectation values [10]. The finite dynamical regime is characterized by sweeping λ and identifying the optimum $\lambda^\star$ that maximizes the dynamical EDT score.

S7. NV XY8 Model

Quantum sensing in NV-center ensembles is analyzed within a quantum resource engineering framework using a rate-based model [11, 12, 13]. For XY8 dynamical-decoupling sequences, the control parameter set is

$$\mathcal{C} = \{P_{\mathrm{laser}},\, t_{\mathrm{read}},\, \Omega_{\mathrm{MW}},\, \tau,\, N_\pi,\, B_0,\, T,\, \mathrm{strain}\}, \tag{14}$$

which determines initialization, control, and environmental conditions of the sensor [14]. These controls map onto an effective resource vector

$$\mathcal{R} = \{R_{\mathrm{ph}},\, C,\, T_2(N_\pi),\, \chi(\omega),\, \mathrm{duty}\}. \tag{15}$$

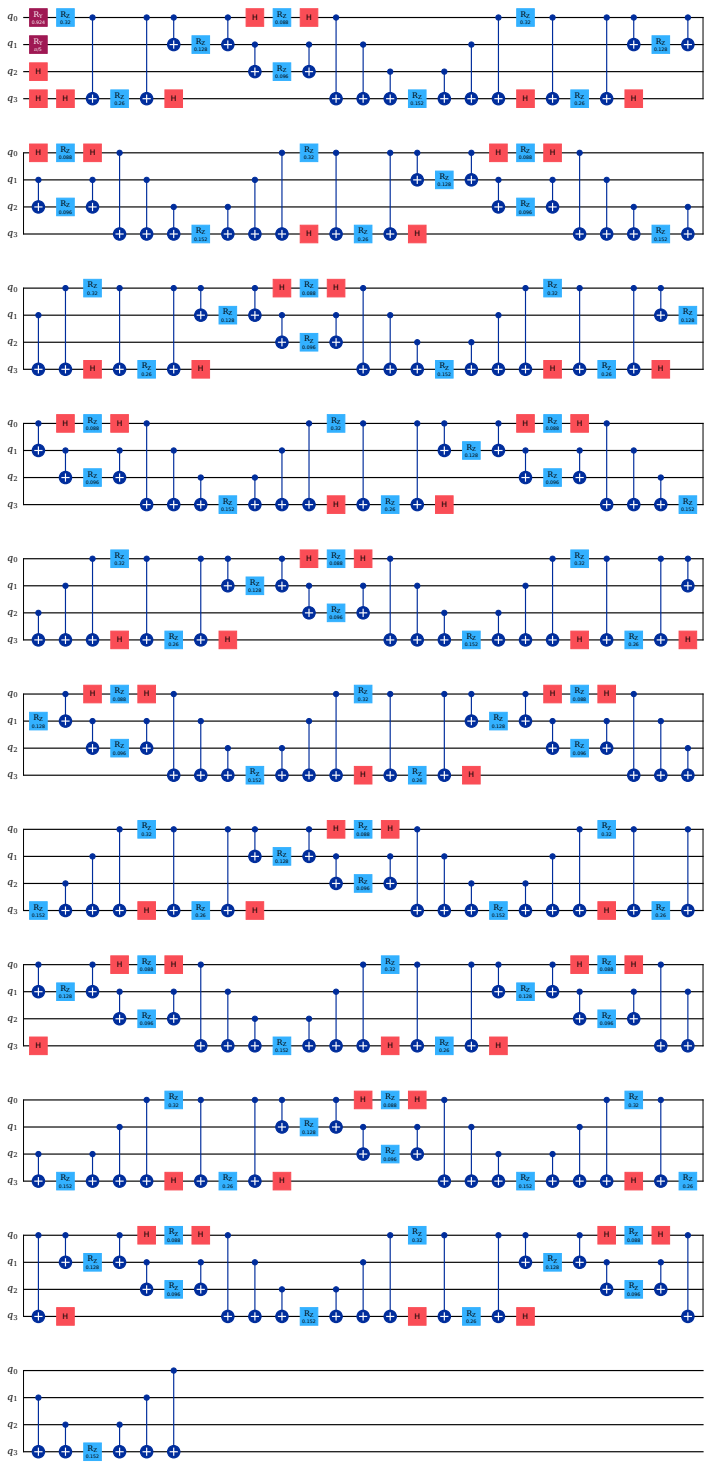


Figure S13: **4-qubit plaquette Trotter realization of EDT dynamics.** Representative first-order Trotter circuit for an archived plaquette-style workflow. The plaquette unitary $e^{-i\lambda Z_0 Z_1 Z_2 Z_3}$ generates multi-qubit entanglement associated with loop topology.

Here R_{ph} is the detected photon rate, C is fluorescence contrast, $T_2(N_\pi)$ is the coherence time under dynamical decoupling, $\chi(\omega)$ is the filter-function response, and duty accounts for initialization and readout overhead. The filter function is peaked near $\omega \simeq \pi/\tau$, enabling frequency-selective amplification of ac fields [15].

The sensitivity proxy is taken to scale as

$$\eta_{\mathrm{AC}}(\omega) \propto \frac{1}{\chi(\omega) C \sqrt{R_{\mathrm{ph}} T_2(N_\pi)}} \times f_{\mathrm{duty}}, \tag{16}$$

consistent with standard analyses of NV magnetometry sensitivity [12, 13]. Parameter sweeps over (τ, N_π) identify finite optimal regimes set by the competition between filter gain, coherence, pulse errors, and timing overhead.

Summary

Across the systems studied here, curvature arising from electron–vibration interaction defines structured manifolds in which electronic states are organized and degeneracies are lifted. The combined supplement documents the normal-mode curvature calculation, the K_3C_{60} finite-window algorithm, the expanded curvature-manifold materials study, the plaquette-model implementation, and the NV XY8 sensing model used to support the main-v4 manuscript.

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